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EE321: Computer Architecture
Control Solution (1h10)
2025 – 2026

Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.

1. Z80 CPU Function (13,5 pts)

Consider Z80 CPU with the following specifications: little-endian 8-bit processor, 16-bit address bus, a partial list of instructions and opcodes is given in Table 1, the instruction formats are given in Table 4. The programming model is given in Table 2 and the GPR codes are given in Table 3. The number of clock cycles for each Z80 machine cycle are given in Table 5. Consider the following HEX memory content:
2000H: DD 21 00 30 DD 7E 00 3C DD 77 00 76 2C D9 2E F8
3000H: F0 01 F2 03 F4 05 F6 07 F8 09 FA 0B FC 0D FD 0F

1.1 Give the size of the registers H, L, IX, and SP.

a) H and L: 8 bits.
b) IX and SP: 16 bits *(1,5 pt.)*

1.2 Which instructions are: a) data movement instructions, and b) control flow instructions?

a) Data movement instructions: `LD`, `PUSH`, and `POP` instructions.
b) Control flow instructions: `JR` and `HALT` instructions. *(1,5 pt.)*

1.3 Which of the states given in Table 6 are executed for: a) `LD IX,nn`, b) `LD IX,(nn)`, and c) `INC (HL)`.

a) IAC, IF, IOD
b) IAC, IF, IOD, OAC, OF
c) IAC, IF, IOD, OAC, OF, DO, OS *(1,5 pt.)*

1.4 Give the size in bytes of the folowing instructions: a) `JR Z,-5`, b) `LD IX,F1H`, and c) `LD A,(IX+5)`.

a) `JR Z,-5`: 2 bytes.
b) `LD IX,F1H`: 4 bytes
c) `LD A,(IX+5)`: 3 bytes. *(1,5 pt.)*

1.5 Give the HEX code of the following instructions: a) `JR Z,-5`, b) `LD IX,F1H`, and c) `LD A,(IX+5)`.

a) `JR Z,-5`: 28FBH
b) `LD IX,(F1H)`: DD21F100H.
c) `LD A,(IX+5)`: DD7E05H *(1,5 pt.)*

1.6 Give the HEX content of the registers PC, MAR, and MBR after executing the **fetch** cycle of the instruction `JR FEH` stored at address 2509H.

The HEX code of the instruction is: 28FEH stored at address 2500H. After the opcode fetch machine cycle, we have:

PC = 250BH
MAR = 250AH
MBR = FEH *(1,5 pt.)*

1.7 Give the HEX content of the registers PC, MAR, and MBR after executing the **execute** cycle of the instruction `JR FEH` stored at address 2509H.

The HEX code of the instruction is: DD2A4823H stored at address 2100H. After the execute cycle, we have:
PC = PC+FE = 250BH - 2 = 2509H
MAR = 250AH
MBR = FEH *(1,5 pt.)*

1.8 Assuming SP=3009H, give the content of registers H, L, SP, and MAR after executing the instruction `POP HL` stored at address 2509H.

HL = [SP] = [3009H] = FA09H
H = FAH
L = 09H
MAR = 300AH
SP = SP + 2 = 300BH *(1,5 pt.)*

1.9 Give the size in bytes and the number of instructions of the program stored at address 2000H.

Size: 12 bytes.
Number of instructions: 5 instructions. *(1,5 pt.)*

2. Basic Concepts (7.5 pts)

2.1 Using unsigned 8-bit representation, give in HEX the result of the following HEX operations: a) 53H - 35H and b) 35H – 53H.

a) 53H – 35H = 1EH
b) 35H – 53H = Can't do this operation *(1,5 pt.)*

2.2 Consider an 8-bit CPU, give in HEX the result of the following HEX operations: a) 53H - 35H and b) 35H – 53H.

c) 53H – 35H = 53H + CBH = 1EH
d) 35H – 53H = 35H + ADH = E2H *(1,5pt.)*

2.3 Consider an 8-bit CPU, give in decimal the result of the following HEX operations: a) 53H - 35H and b) 35H – 53H.

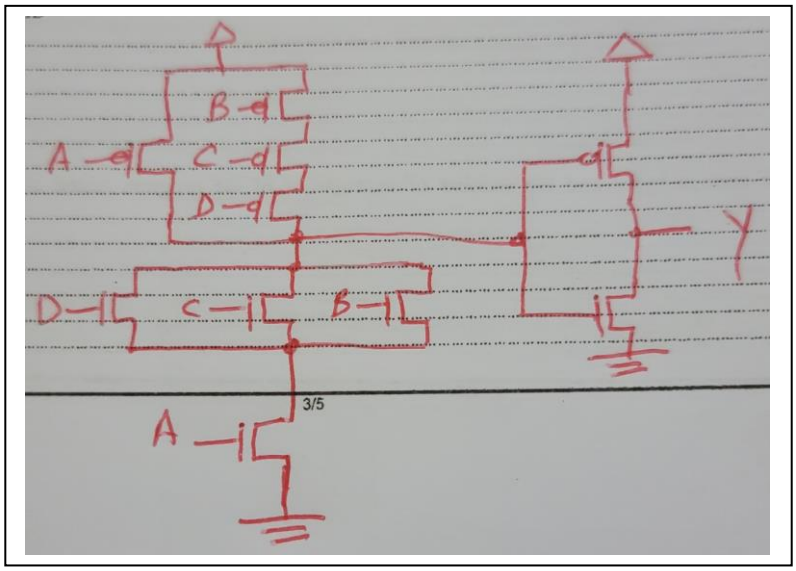
a) 53H – 35H = 1EH = NaN
b) 35H – 53H = 0FH = NaN *(1,5 pt.)*

2.4 Consider an 8-bit CPU, which of the carry and overflow flags are affected by following HEX operations: a) 53H - 35H and b) 35H – 53H.

a) None
b) Carry flags *(1,5 pt.)*

2.5 Draw the CMOS transistor schematic for the gate that implements the function: $y = AB + AC + AD$

$y = AB + AC + AD = A(B + C + D)$



(1,5 pts.)

Instruction	Opcode (binary or HEX)	Instruction Size	Description
LD R, S	$01R_2R_1R_0\ S_2S_1S_0$		R= S
LD IX, nn	DD21H		IX = nn
LD IX, (nn)	DD2AH		IX = [nn]
LD HL, nn	21H		HL = nn
LD HL, (nn)	2AH		HL = [nn]
LD R, (IX+n)	$11011101\ 01\ R_2R_1R_0110$		R = [IX+n]
LD (IX+n), S	$11011101\ 01110\ S_2S_1S_0$		[IX+n] = S
LD (nn), A	32H		[nn] = A
PUSH IX	DDE5H		SP = SP-2, [SP] = IX
POP IX	DDE1H		IX=[SP], SP = SP+2
PUSH HL	E5H		SP = SP-2, [SP] = HL
POP HL	E1H		HL=[SP], SP = SP+2
JR n	18H		PC =PC + n
JR Z,n	28H		IF zero flag, PC = PC + n
ADD A, n	C6H		A = A + n
ADD A, (HL)	86H		A = A + [HL]
ADD A, S	$10000S_2S_1S_0$		A = A + S
INC (HL)	34H		[HL] = [HL] + 1
INC R	$00R_2R_1R_0100$		R = R + 1
NOP	00H		No operation
HALT	76H		Stop the execution of the program

R =Destination register (it can be accumulator or any GPR register).
S = Source register (it can be accumulator any GPR register).
$R_2R_1R_0$ = Binary code of the register R (see Table 3).
$S_2S_1S_0$ = Binary code of the register S (see Table 3).
n =two's complement 8-bit value.
nn =16-bit value.
[…] = Content of memory location.

Table 1: Partial list of Z80 instructions.

Register names	Description
A	Accumulator
B, C, D, E, H, and L	General purpose registers (GPR), can also be used in pairs as 16-bit registers (i.e. BC, DE, HL)
IX, IY	Address registers
F	8-bit flag register
I	8-bit interrupt register
R	8-bit refresh register
PC	Program counter
SP	Stack pointer

Table 2: Z80 programming model.

GPR	A	B	C	D	E	H	L	
Code	111	000	001	010	011	100	101	

Table 3: Binary codes of Z80 accumulator and GPR.

1-byte
Opcode

1-byte instruction format

1-byte	1-byte
Opcode	Operand n or Opcode

2-bytes instruction format.

1-byte	2-bytes
Opcode	Operand nn or (Opcode + Operand n)

3-bytes instruction format.

2-bytes	2-bytes
Opcode	Operand nn

4-bytes instruction format

Table 4: Z80 instruction formats.

Z80 machine cycle	Number of clock cycles (T-states)
1-byte opcode fetch	4 clock cycles
1-byte operand fetch	3 clock cycles
1-byte memory read	3 clock cycles
1-byte memory write	3 clock cycles
1-byte I/O read	4 clock cycles
1-byte I/O write	4 clock cycles

Table 5: Number of clock cycle for each Z80 machine cycle.

State	Definition
IAC	Instruction Address Calculation
IF	Instruction Fetch
IOD	Instruction Operation Decoding
OAC	Operand Address Calculation
OF	Operand Fetch
DO	Data Operation
OS	Operand Store

Table 6: Instruction cycle states definition.